# Devixo Solutions

### AI/ML Internship Program

# Task 01: Python Programming, Data Analysis & Visualization

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| **Internship Domain** | AI/ML |
| **Date** | 11/10/2026 |

# 1. Introduction

This project applies Python programming and data-analysis techniques to a real-world e-commerce dataset from Pakistan. The work covers data loading, cleaning, exploratory data analysis (EDA), statistical summaries, and ten visualizations to examine transaction-record patterns, product categories, payment methods, prices, and monthly activity.

### Project objectives

• Load and inspect a large dataset.
• Identify empty rows and columns, assess missing values, and convert date fields.
• Summarize numerical and categorical variables.
• Create ten required visualization types.
• Derive evidence-based insights and document limitations.

# 2. Dataset Overview

| Attribute | Description |
|---|---|
| Dataset | Pakistan's Largest E-Commerce Dataset |
| Source | Kaggle |
| Source link | Dataset_Kaggle_Link |
| Initial shape | 1,048,575 rows × 26 columns |
| Completely empty rows removed | 464,051 |
| Completely empty columns removed | 5 |
| Cleaned working shape | 584,524 rows × 21 columns |
| Transaction date range | 2016-07-01 to 2018-08-28 |
| Unique customers | 115,326 |
| Valid product categories | 15 (excluding missing-marker values) |

# 3. Methodology and Tools

The analysis was carried out in Google Colab using Python. Pandas was used to load, inspect, clean, group, and summarize the data; NumPy supported numerical operations; Matplotlib and Seaborn were used to create charts.

### Workflow

1. Loaded the CSV dataset into a Pandas DataFrame.
2. Inspected the first, last, and random rows, column names, data types, shape, and summary information.
3. Removed fully empty rows and columns and stripped surrounding whitespace from column names.
4. Checked duplicate rows and missing values.
5. Replaced missing sales commission codes with “Unknown” and retained records with missing product categories in the main cleaned dataset, using category-valid subsets for category plots.
6. Converted transaction and working-date columns to datetime and parsed the customer-since field as a month-based date.
7. Calculated summary statistics, category/payment/status counts, and monthly record counts.
8. Produced ten visualizations and documented findings.

# 4. Data Cleaning Results

| Quality check | Result / treatment |
|---|---|
| Original dataset | 1,048,575 rows × 26 columns |
| Completely empty rows | 464,051 removed |
| Completely empty columns | 5 removed |
| Exact duplicate rows | 0 found |
| Missing sales commission codes | 137,178 filled as “Unknown” |
| Missing product categories | 164 actual nulls; excluded from category-only analysis |
| Zero-price records | 2,232 retained and flagged for review |
| Negative grand totals | 76 retained and flagged for review |
| Negative discounts | 3 observed and retained for review |
| Quantities above 100 | 42 observed and retained for review |
| Final working dataset | 584,524 rows × 21 columns |

# 5. Exploratory Data Analysis

The following results summarize the cleaned working dataset. Numerical values are rounded where appropriate.

| Measure | Result |
|---|---|
| Transaction records | 584,524 |
| Unique customers | 115,326 |
| Valid product categories | 15 |
| Average item price | 6,348.75 |
| Median item price | 899.00 |
| Average quantity per record | 1.30 |
| Most common category | Mobiles & Tablets (115,710 records) |
| Most common payment method | COD (46.53% of records) |
| Most common status | complete (233,685 records) |
| Peak month by record count | November 2017 (83,928 records) |
| Zero-price records | 2,232 |
| Negative grand totals | 76 |

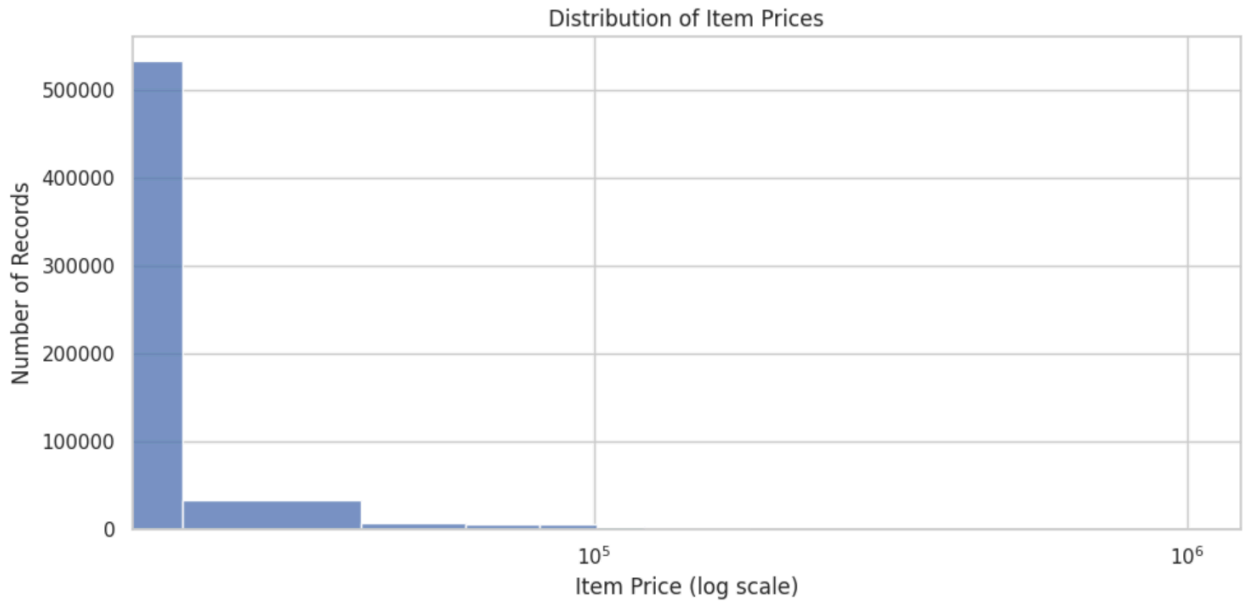
The difference between mean item price (6,348.75) and median item price (899) indicates a right-skewed distribution, where relatively expensive items raise the average. Monthly counts are counts of transaction records, not confirmed revenue or unique orders.

# 6. Data Visualizations

All ten chart screenshots provided for this report are included below with a short interpretation.

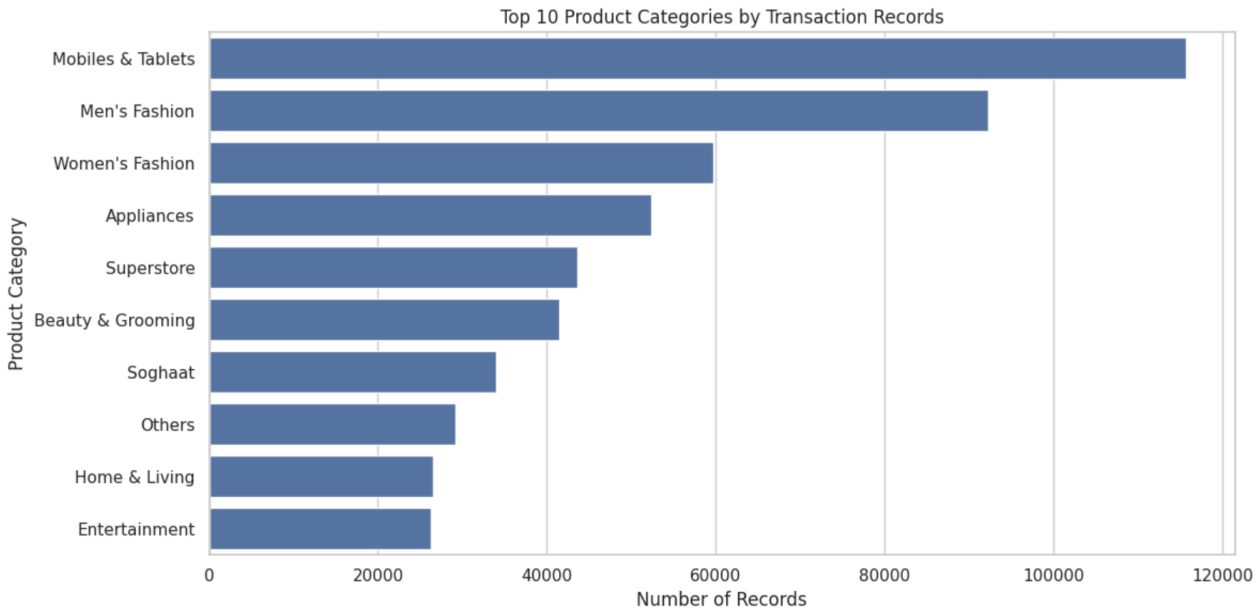
## Chart 1. Histogram: Distribution of Item Prices

The price distribution is strongly right-skewed: most records are concentrated at lower prices, while a smaller number of expensive items extend the distribution. A logarithmic x-axis helps display the wide price range.



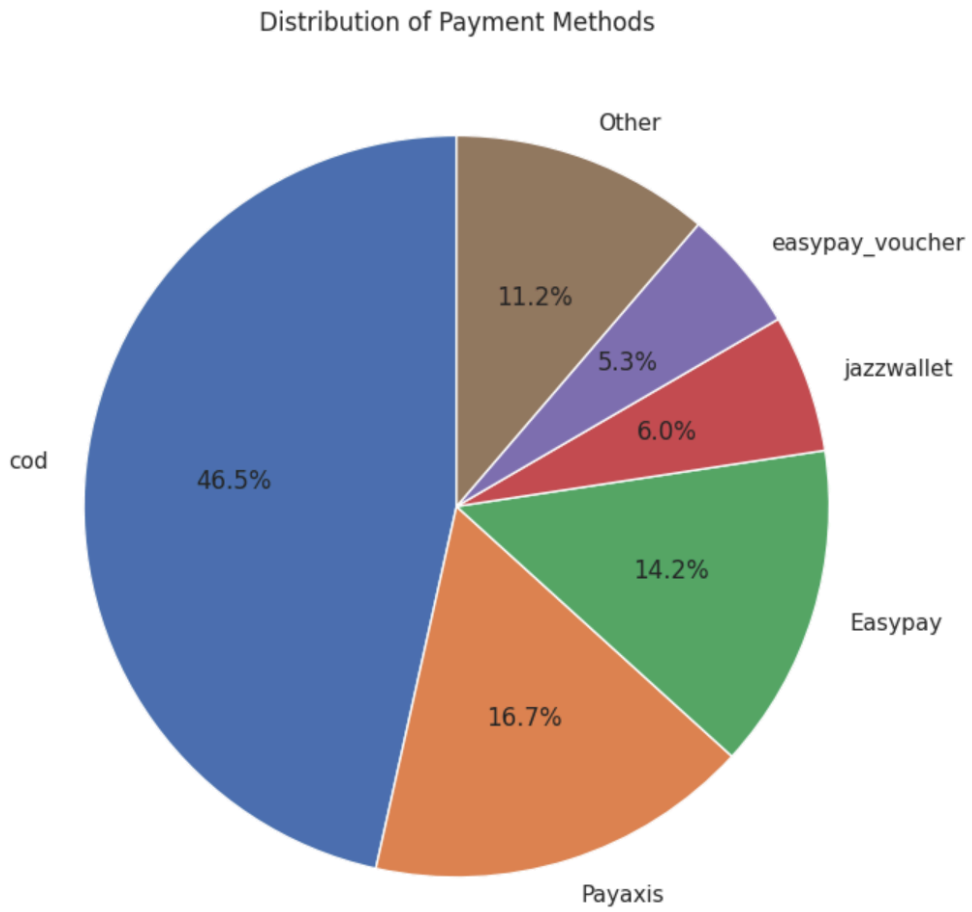

## Chart 2. Bar Chart: Top 10 Product Categories

Mobiles & Tablets is the most frequently recorded category, followed by Men's Fashion and Women's Fashion. These counts describe transaction records, not necessarily unique orders.



## Chart 3. Pie Chart: Payment Method Distribution

Cash on Delivery (COD) is the most common payment method, representing approximately 46.53% of all records. The chart groups less common methods into 'Other' for readability.



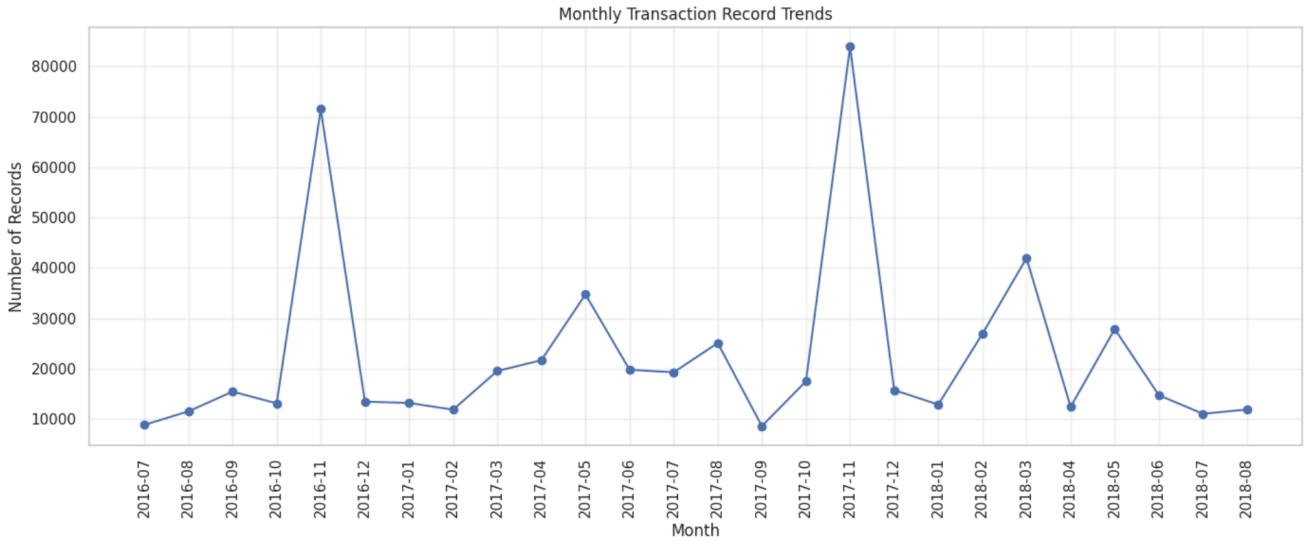

## Chart 4. Scatter Plot: Item Price vs. Quantity Ordered

The scatter plot explores whether item price and ordered quantity move together. A visual pattern may suggest an association, but the chart alone cannot establish causation.



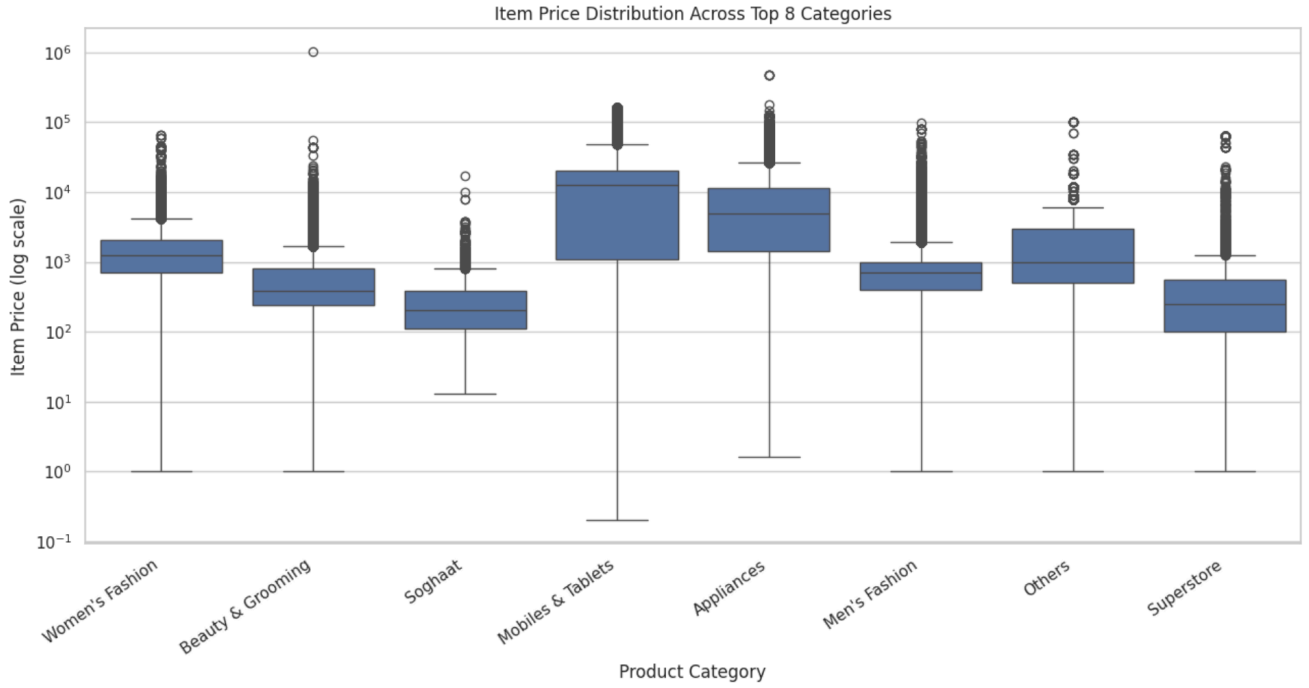
## Chart 5. Line Chart: Monthly Transaction Record Trends

November 2017 is the peak month in the cleaned records, with 83,928 records. November 2016 also shows a notable spike; seasonal explanations would require additional investigation.



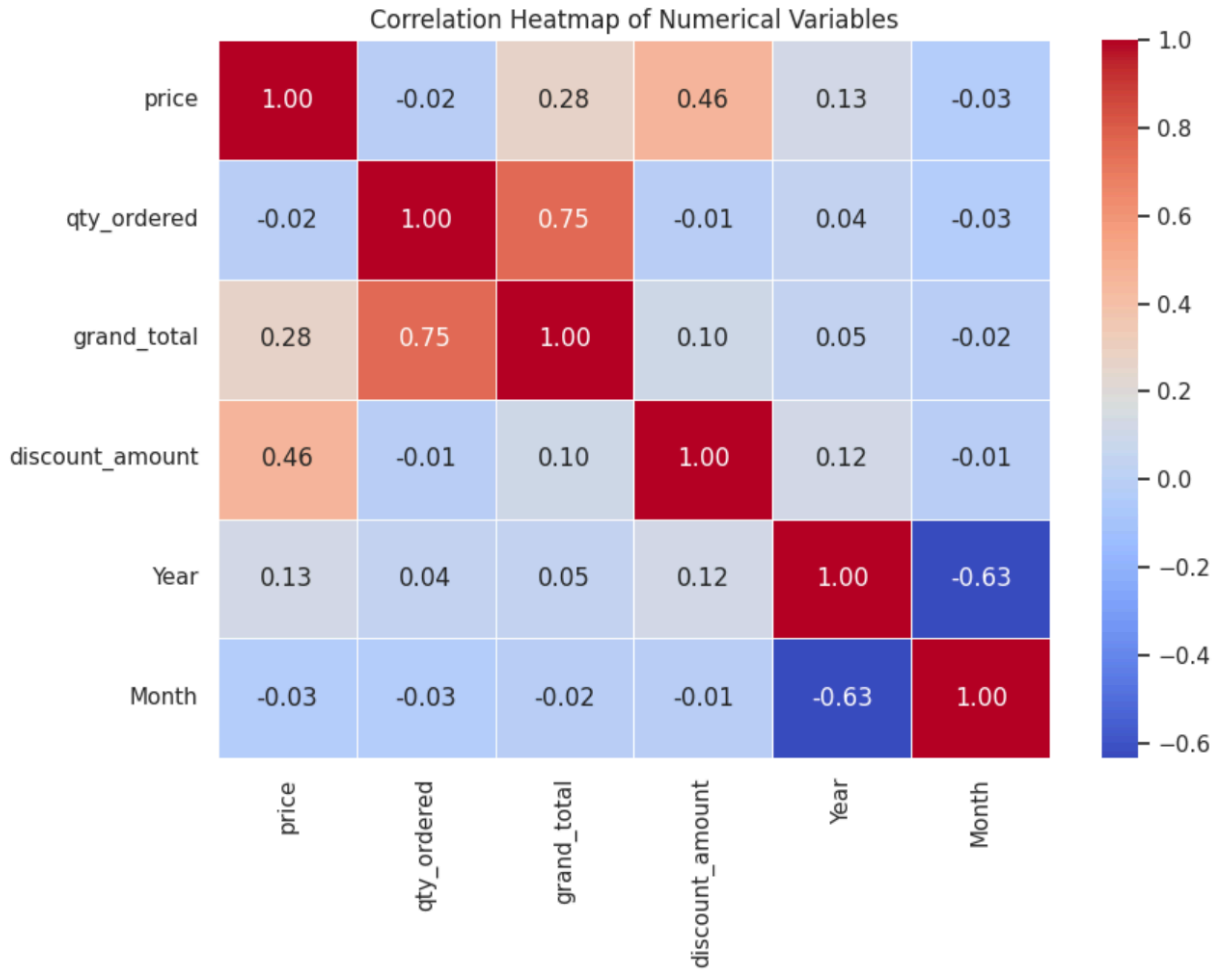

## Chart 6. Box Plot: Price Distribution by Category

The box plot compares price spread across the eight most common valid categories. Points or whiskers far from the central range may indicate outliers; they are not automatically data errors.



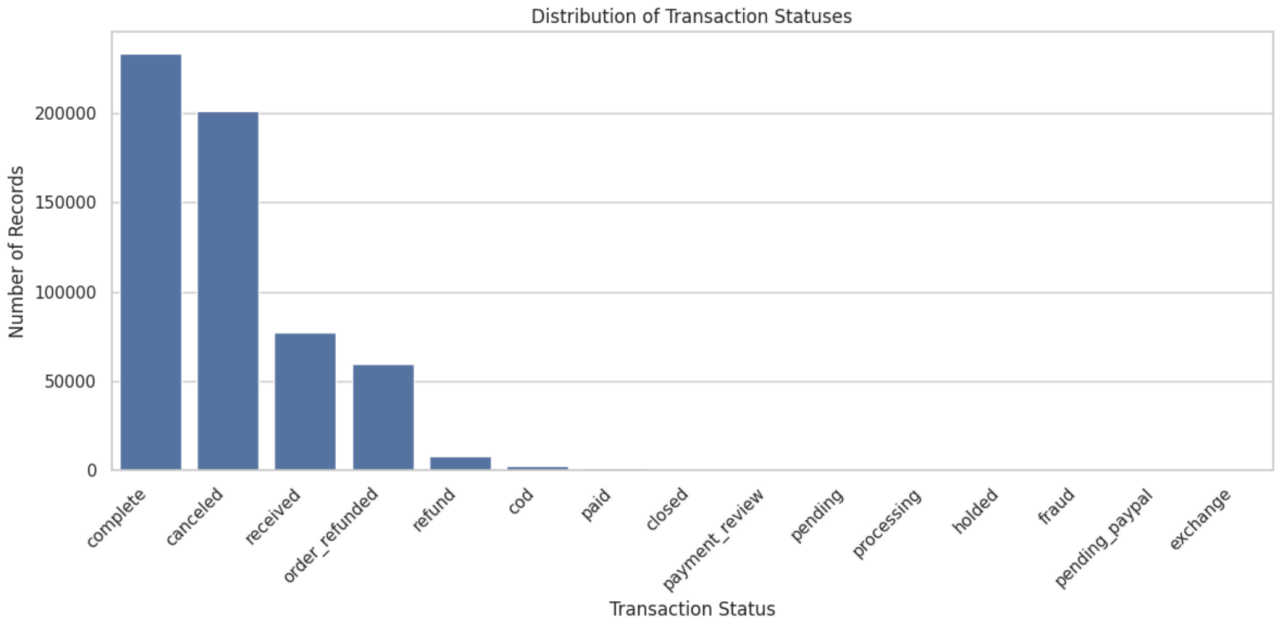

## Chart 7. Correlation Heatmap: Numerical Variables

The heatmap summarizes pairwise correlations among selected numerical variables. Correlations describe linear association and do not prove causation; calendar variables should be interpreted cautiously.




## Chart 8. Count Plot: Transaction Statuses

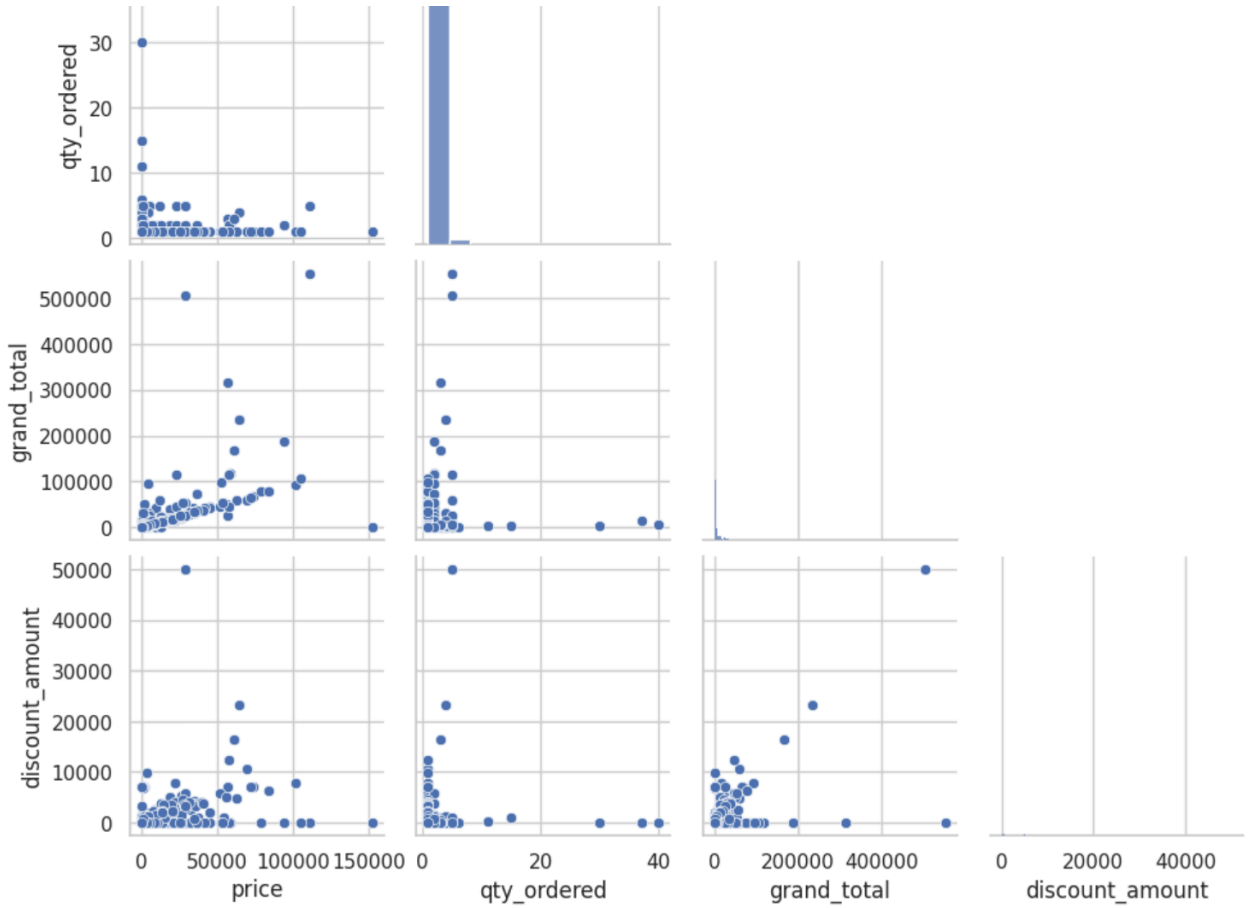
The most frequent status is 'complete', followed by 'canceled'. These are status-record counts and should not be interpreted directly as a cancellation rate or unique-order totals.




## Chart 9. Pair Plot: Numerical Variable Relationships

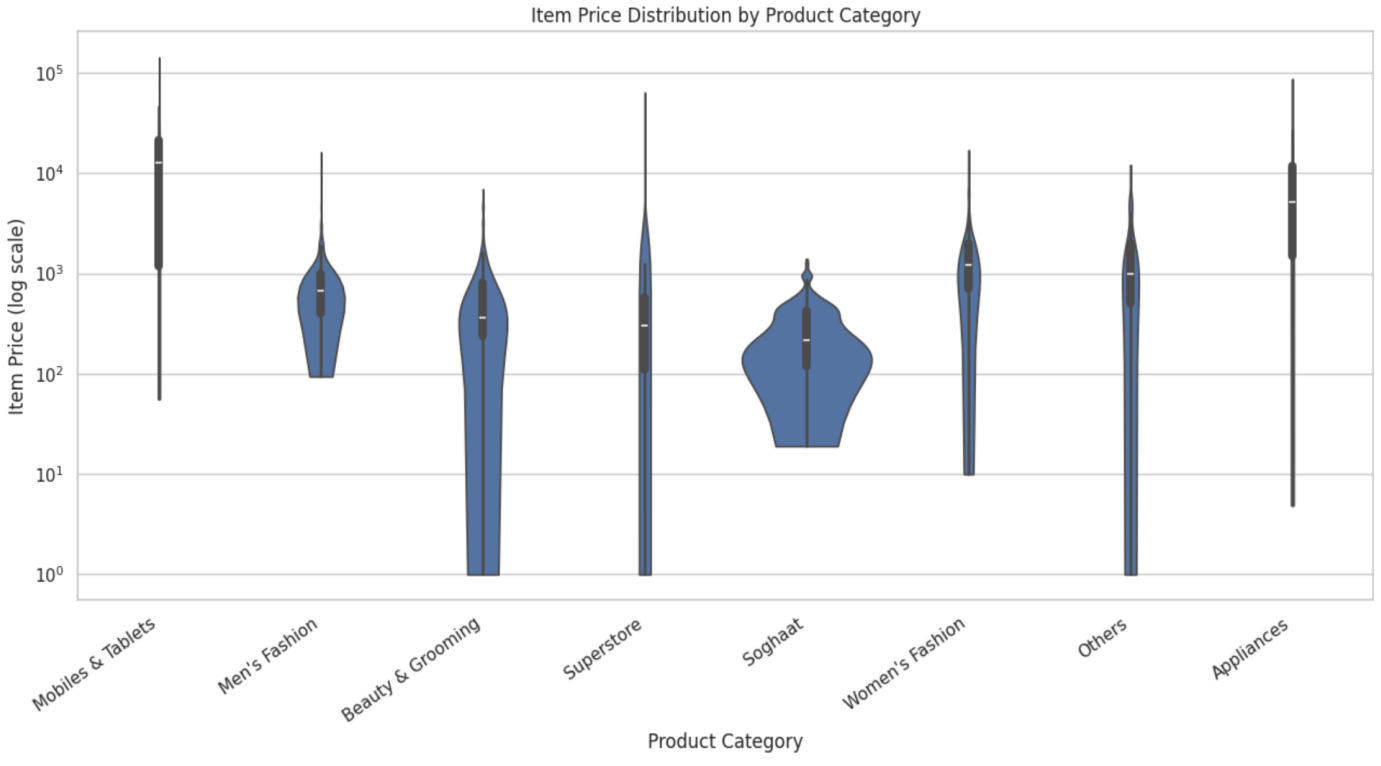
The pair plot compares distributions and pairwise patterns for price, quantity, grand total, and discount amount using a

sample of records to keep the visualization manageable.




## Chart 10. Violin Plot: Price Distribution by Category

The violin plot compares the shape and density of prices across selected categories. Wider areas indicate ranges with more observations; the logarithmic price scale makes differences easier to inspect.



# 7. Key Insights and Findings

1. **Dataset scale:** The cleaned working dataset contains 584,524 transaction records, providing a substantial basis for exploratory analysis.

2. **Customer base:** There are 115,326 unique customer IDs represented in the dataset.

3. **Popular category:** Mobiles & Tablets is the most frequently recorded valid category, with 115,710 records. 4.

**Payment preference:** COD is the most common payment method, representing approximately 46.53% of records.

5. **Skewed prices:** The average item price (6,348.75) is much higher than the median (899), suggesting that high-priced items influence the mean.

6. **Status distribution:** “complete” is the most common recorded status with 233,685 records, followed by “canceled” with 201,249 records.

7. **Monthly activity:** November 2017 is the peak month by record count, with 83,928 records. This spike may warrant further investigation, but the reason cannot be established from counts alone.

8. **Ordered quantity:** The average quantity per record is about 1.30, suggesting most records have relatively small quantities, while some large quantities are present.

9. **Category price differences:** In the category summary, Entertainment has the highest median item price (19,275) and Books the lowest (150), excluding the missing-marker category.

10. **Data quality:** The data contains 2,232 zero-price records and 76 negative grand totals. These records should be investigated in context rather than removed automatically.

# 8. Conclusion

This project analyzed Pakistan's Largest E-Commerce Dataset using Python, Pandas, NumPy, Matplotlib, and Seaborn. After removing completely empty rows and columns and addressing selected missing values, the working dataset contained 584,524 transaction records and 115,326 unique customers across 15 valid product categories.

Exploratory data analysis revealed that Mobiles & Tablets was the most frequently recorded category, while Cash on Delivery was the most commonly used payment method, accounting for approximately 46.53% of records. The average item price was 6,348.75, compared with a median of 899, indicating a right-skewed price distribution. November 2017 recorded the highest monthly activity, with 83,928 transaction records.

Ten visualizations were created to explore price distributions, category popularity, payment methods, monthly trends, numerical relationships, and transaction statuses. The analysis also identified 2,232 zero-price records and 76 records with negative grand totals, highlighting areas that may require further investigation.

Overall, this project strengthened practical skills in data cleaning, exploratory data analysis, statistical interpretation, and data visualization. The findings provide a useful overview of customer activity and transaction patterns, although further investigation would be needed to establish causes or make definitive business-performance conclusions.

# 9. References

1. Kaggle. Pakistan's Largest E-Commerce Dataset. https://www.kaggle.com/datasets/zusmani/pakistans-largest-ecommerce-dataset

2. Python Software Foundation. Python Documentation. https://docs.python.org/3/

3. pandas development team. pandas documentation. https://pandas.pydata.org/docs/

4. Matplotlib development team. Matplotlib documentation. https://matplotlib.org/stable/

5. Seaborn development team. Seaborn documentation. https://seaborn.pydata.org/